

Java Lambda Expressions

```
import java.io.*;
import java.util.*;
interface PerformOperation {
    boolean check(int a);
}
class MyMath {
    public static boolean checker(PerformOperation p, int num) {
        return p.check(num);
    }
}
PerformOperation isOdd() {
    return n -> n % 2 != 0;
}
PerformOperation isPrime() {
    return n -> {
        if (n < 2) return false;
        for (int i = 2; i * i <= n; i++) {
            if (n % i == 0)
                return false;
        }
        return true;
    };
}
PerformOperation isPalindrome() {
    return n -> {
        int temp = n, rev = 0;
        while (temp > 0) {
            rev = rev * 10 + temp % 10;
            temp /= 10;
        }
        return rev == n;
    };
}
}
public class Solution {
    public static void main(String[] args) throws IOException {
        MyMath ob = new MyMath();
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
        int T = Integer.parseInt(br.readLine());
        PerformOperation op;
        boolean ret = false;
        String ans = null;
        while (T-- > 0) {
            String s = br.readLine().trim();
            StringTokenizer st = new StringTokenizer(s);
            int ch = Integer.parseInt(st.nextToken());
            int num = Integer.parseInt(st.nextToken());
            if (ch == 1) {
                op = ob.isOdd();
                ret = ob.checker(op, num);
                ans = (ret) ? "ODD" : "EVEN";
            } else if (ch == 2) {
                op = ob.isPrime();
                ret = ob.checker(op, num);
                ans = (ret) ? "PRIME" : "COMPOSITE";
            } else if (ch == 3) {
```

```

op = ob.isPalindrome();
ret = ob.checker(op, num);
ans = (ret) ? "PALINDROME" : "NOT PALINDROME";
}
System.out.println(ans);
}
}
}

```

✓ Test case 0

✓ Test case 1 

✓ Test case 2 

1	5
2	1 4
3	2 5
4	3 898
5	1 3
6	2 12

Expected Output

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1	EVEN
2	PRIME
3	PALINDROME
4	ODD
5	COMPOSITE

```

import java.util.*;
class Checker implements Comparator<Player> {
    public int compare(Player a, Player b) {
        if (a.score != b.score) {
            return b.score - a.score;
        }
        return a.name.compareTo(b.name);
    }
}
class Player{
    String name;
    int score;
    Player(String name, int score){
        this.name = name;
        this.score = score;
    }
}

```

```

class Solution {
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);
        int n = scan.nextInt();
        Player[] player = new Player[n];
        Checker checker = new Checker();
        for(int i = 0; i < n; i++){
            player[i] = new Player(scan.next(), scan.nextInt());
        }
        scan.close();
        Arrays.sort(player, checker);
        for(int i = 0; i < player.length; i++){
            System.out.printf("%s %s\n", player[i].name, player[i].score);
        }
    }
}

```

✓ Test case 0

✓ Test case 1 

✓ Test case 2 

✓ Test case 3 

✓ Test case 4 

✓ Test case 5 

Compiler Message

Success

Input (stdin)

```

1 5
2 amy 100
3 david 100
4 heraldo 50
5 aakansha 75
6 aleksa 150

```

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Expected Output

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